

Hypothesis Testing I

Unit - 3 Session - 6

Unit 3: Estimation & Hypothesis Testing

- Sampling Distribution & Central limit theorem,
- Point estimate
- confidence interval
- Hypothesis Testing Basics
- Null & Alternative Hypothesis formulation - Types of hypotheses
- Hypothesis testing process
- Errors in hypothesis testing - Type I and Type II
- Power of test
- Number of tails
- Choice of test statistic

Testing of Hypothesis



Question:

A manufacturer produces batteries. He claims that the life of his batteries is 25,500 hours. Can we verify his claim?



Solution:

A trivial solution is to obtain the life of all the batteries produced by the manufacturer by using them and thereafter verify this claim

However, by doing so all the manufactured products (batteries) will be exhausted and he will not have any batteries to sell.

To avoid such scenarios, we **statistically test the hypothesis**

What is a hypothesis?

A hypothesis is a claim about the population parameter

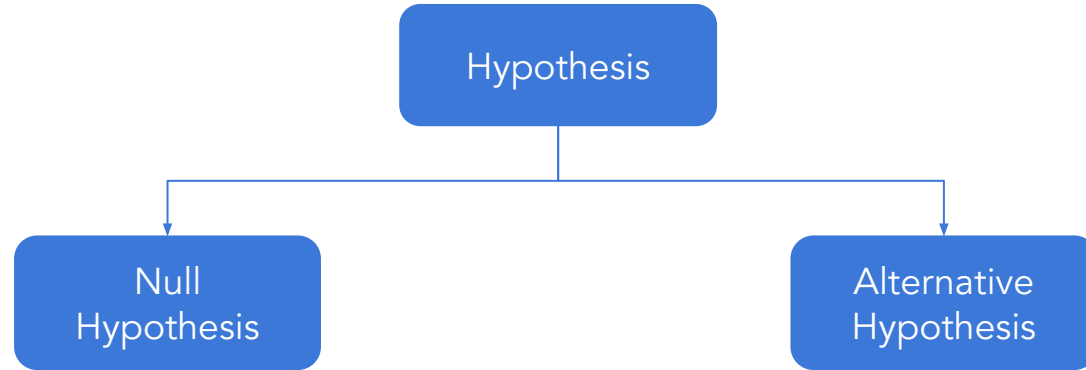
Examples:

- The average diameter of bolts manufactured is 200 cm
- The new drug is more effective

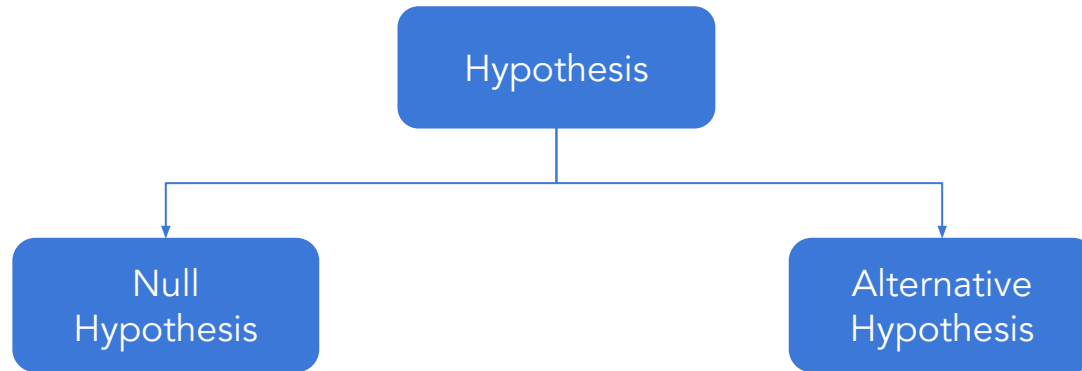
Hypothesis - example

A pharmaceutical has claimed to have produced a new drug is effective against cancer treatment.

To test the pharmaceuticals claim, first write the test hypotheses -
Null Hypothesis and the Alternative Hypothesis



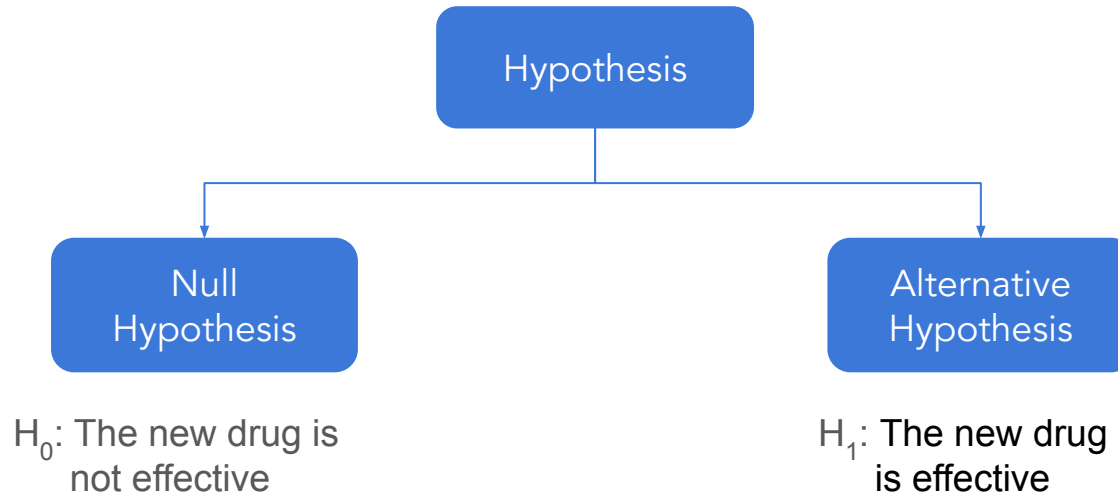
Null and alternative hypothesis



- Hypothesis of no change
- Denoted by H_0
- It directly contradicts the null hypothesis
- Also known as research hypothesis
- Denoted by H_a or H_1

Hypothesis - example

A pharmaceutical has claimed to have produced a new drug which treats the patients against cancer.



Hypothesis - example

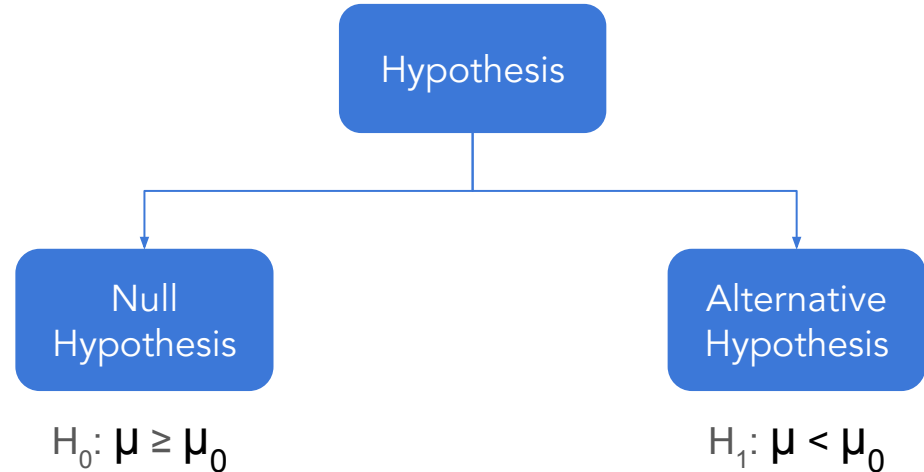
- Now consider the effectiveness of the new drug is determined by the number of days a patient requires to recover.
- The data for the number of days the patient take to recover are recorded
- Let the random variable X be the number of days required to recover
- Thus μ is the average number of days required to recover

State the hypothesis

It is claimed that the new drug reduces the recovery time

μ : the average number of days required to recover

Here, μ_0 is the general recovery time



Hypothesis testing

- The null and alternative hypothesis are contrasting
- Either the null or the alternative hypothesis can be true
- The decision of the test procedure is one of the following:
 - Reject H_0 implies the null hypothesis is false
 - Fail to reject H_0 implies the null hypothesis is true



“Fail to reject H_0 ”

- The null hypothesis is assumed to be true unless there is sufficient evidence to reject it.(we have insufficient evidence to support H_1)
- Failing to reject H_0 does not confirm that H_0 is true; it only means there is not enough evidence against it.
- The alternative hypothesis has the burden of proof, i.e. from the data it will have enough evidence to say the null hypothesis is false
- Hence, it is preferred to say ‘fail to reject H_0 ’ instead of accept H_0

Errors in Hypothesis Testing

Probability of committing an error

- $P(\text{Type I error})$ is the probability of wrongly rejecting a true null hypothesis
- It is the level of significance (α) set for the hypothesis
- An α of 0.05 suggests that one is willing to accept a 5% chance that you reject a true null hypothesis
- In order to lower this risk, set a lower α . However, it will be more difficult to reject the false null hypothesis

Error in hypothesis testing

		Test Decision	
		Reject H_0	Accept H_0
Actual Situation	H_0 is True	Type I error Wrongly reject true H_0	Correct Conclusion Correctly accept true H_0
	H_0 is False	Correct Conclusion Correctly reject false H_0	Type II error Wrongly accept false H_0

Probability of committing an error

- P(Type II error) is the probability of wrongly accepting a false null hypothesis
- Denoted by β
- $1 - \beta$ is the **power of the test**, i.e. correctly rejecting a false null hypothesis
- In order to lower the risk of type II error, ensure the power of the test is high
- In python, we calculate the power of a Z test using the **'power.zt_ind_solve_power()'** from statsmodels library



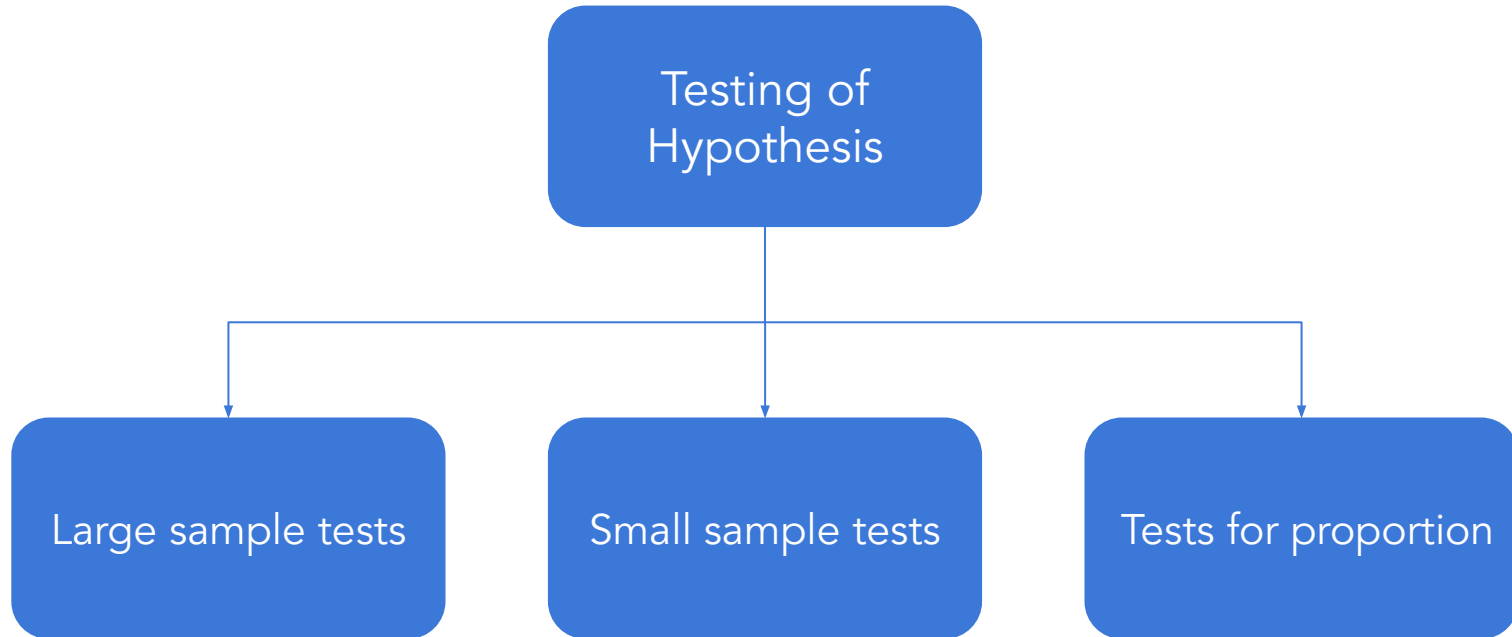
Controlling the errors

- Both errors cannot be controlled simultaneously
- Fix one error and minimize the other
- Generally α is fixed and β is controlled

Previously

- The unknown population parameters are estimated by from sample
- The unknown population parameters are replaced by the sample estimates
- Test the hypothesis for population mean when the population variance is known

Now...



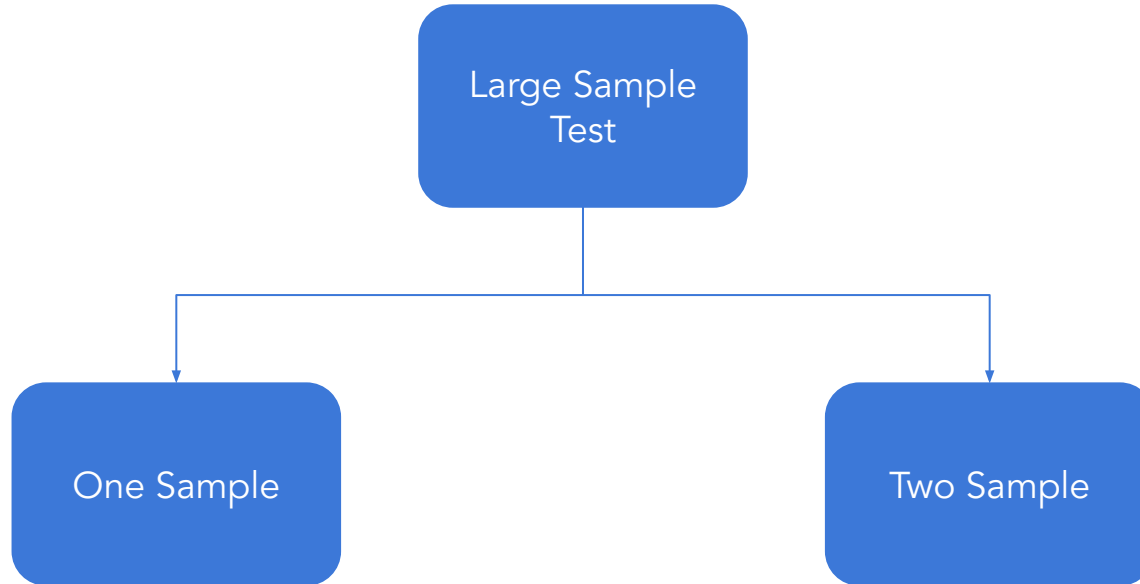
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Large Sample Tests

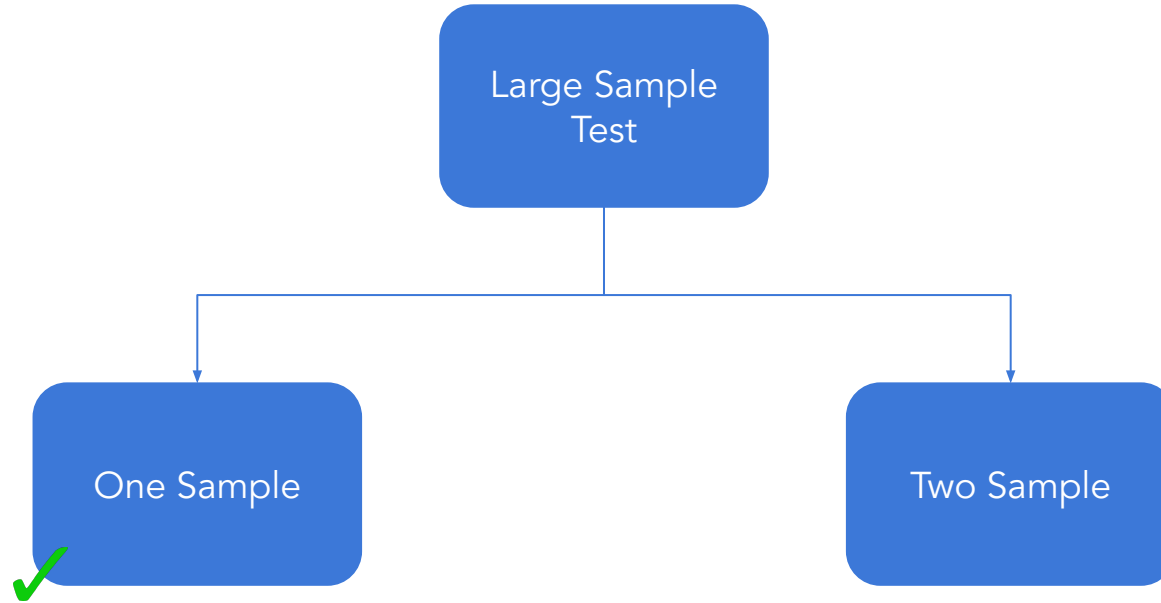
Large sample tests

- For large n , according to CLT, the sampling distribution of mean follows normal distribution
- For tests $n \geq 30$ are considered to be large sample tests
- If σ is unknown, its point estimator - sample standard deviation - is used

Large sample tests



Large sample tests



Large sample tests

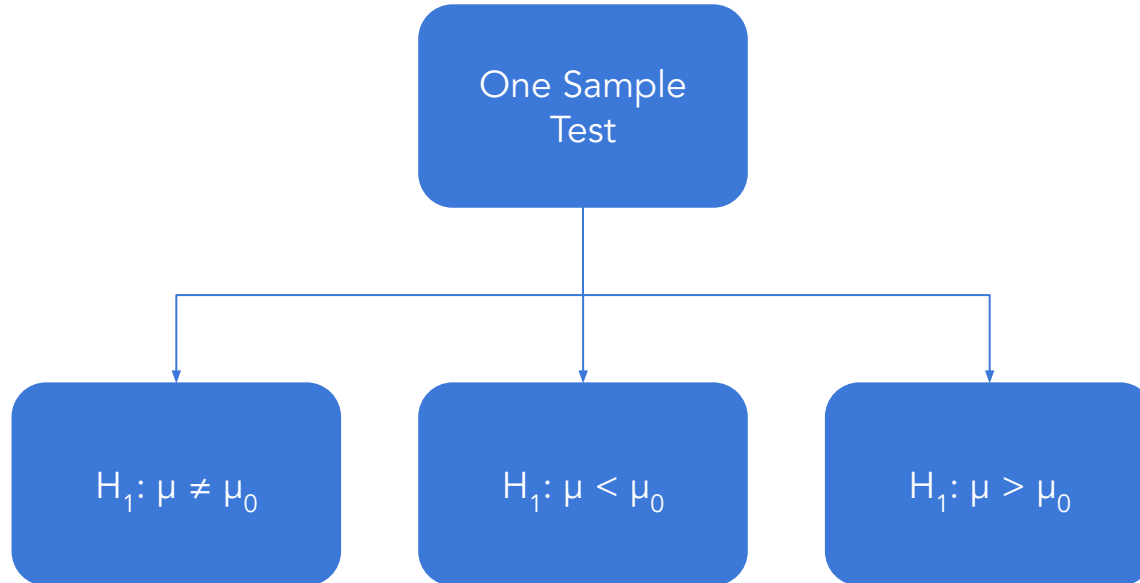
- One sample

- Two sample

Number of Defective Items in a Box
5
8
2
5
3
6

Can we say that each box has less than 3 defective items?

One sample test



One sample tests

- In the last session, we considered the case where σ^2 is known
- The test statistic changes slightly for σ^2 is unknown
- σ^2 replaced by sample variance (s^2)



Question:

Write the null and alternative hypothesis for the following scenarios:

1. A bolt manufacturing company claims that the average number of bolt manufactures in a day is 60
2. The new virus testing kit takes 20 hours less time than usual testing kit that takes 34 hours for the results
3. An analyst wants to test whether the Apple Inc. has outperformed in 2019 by 13.5%

Tests based on Z statistic

- The test statistic Z given by

$$Z = \frac{\bar{X} - \mu}{s / \sqrt{n}}$$

Diagram illustrating the components of the Z statistic formula:

- Sample mean** points to $\bar{X}$
- Specified mean** points to μ
- Sample standard deviation** points to s
- Sample size** points to n

where $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$

- Under H_0 , the test statistic follows normal distribution



The python code to conduct a Z test for population mean is

```
statsmodels.stats.weightstats.ztest(sample, value, alternative)
```



Why n-1?

- The **unbiased estimator** of population variance is $s^2 = \frac{n(\text{sample variance})}{n-1}$
- Also $\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$ and $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$ have the same numerator, thus dividing it by n-1 gives a smaller value which is unbiased
- If a sample of size 1 is drawn, say the observation is 475, the mean would be is 475 and there is no variation with just one observation. Thus, the denominator is n-1 is more appropriate

Write the null and alternative hypothesis and identify the type of the test.

- To test the hypothesis that the mean diastolic blood pressure for a group of 85 adults is less than 90mm
- The tensile strength of an alloy is more than 500 Pa
- The average heart rate of a healthy human is 85 beats per minute
- The yield BT cotton is more than 450 kg lint per hectare



Question:

Write the null and alternative hypothesis for the following scenarios:

1. A bolt manufacturing company claims that the average number of bolt manufactures in a day is 60
2. The new virus testing kit takes 20 hours less time than usual testing kit that takes 34 hours for the results
3. An analyst wants to test whether the Apple Inc. has outperformed in 2019 by 13.5%



1. A bolt manufacturing company claims that the average number of bolts he manufactures in a day is 60

Solution:

The null and alternative hypothesis for the following scenarios:

Let X : the number of bolts manufactured in a day and

μ : the average number of bolts manufactured in a day

$H_0: \mu = 60$ Against $H_1: \mu \neq 60$

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Two tailed test

Two-tailed test is a method in which the **critical region** of a distribution is split in its two tails. It tests whether a sample is greater than or less than the population parameter.

A critical region is a region in which the null hypothesis is rejected. It is also known as 'Rejection Region'.

Example

Let X: the number of bolts manufactured in a day and

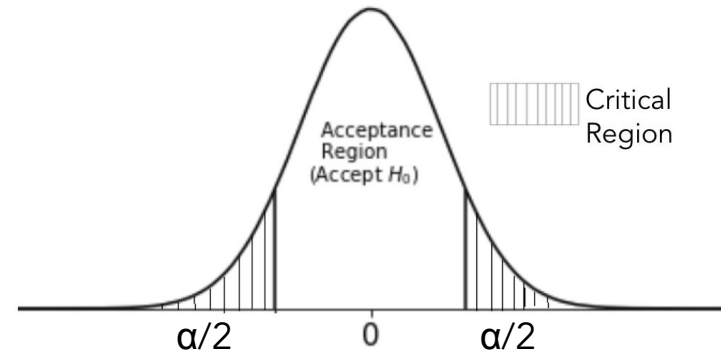
μ : the average number of bolts manufactured in a day

$$H_0: \mu = 60$$

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Against $H_1: \mu \neq 60$

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2. The new virus testing kit takes less time than usual testing kit that takes 34 hours for the results

Solution:

The null and alternative hypothesis for the following scenarios:

Let X: the time required for the results by a testing kit

μ : the average time required for the results by a testing kit

$H_0: \mu \geq 34$ Against $H_1: \mu < 34$

Left tailed test

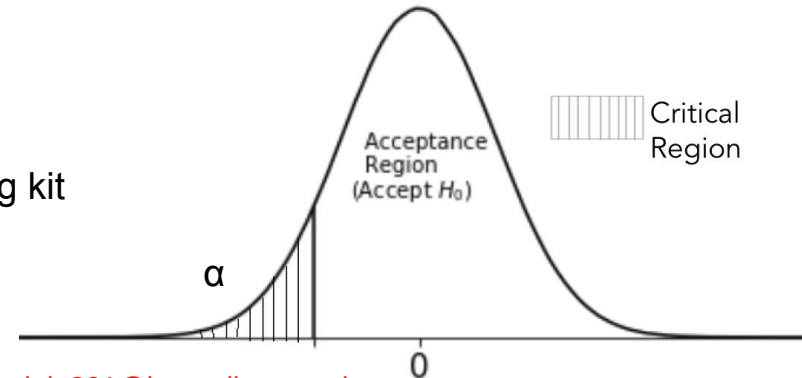
A left-tailed test is a statistical hypothesis test set up to show that the sample parameter would be lower than the population parameter.

Example

Let X: the time required for the results by a testing kit

μ : the average time required for the results by a testing kit

$H_0: \mu \geq 34$ Against $H_1: \mu < 34$





3. An analyst wants to test whether the Apple Inc. has outperformed in 2019 by 13.5%

Solution:

The null and alternative hypothesis for the following scenarios:

Let X: the performance of Apple Inc. stock

μ : the average performance of Apple Inc. stock

$H_0: \mu \leq 13.5$ Against $H_1: \mu > 13.5$

Right tailed test

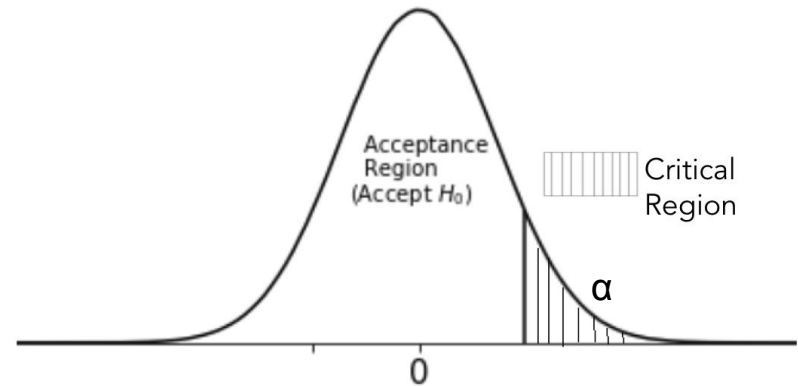
A right-tailed test is a statistical hypothesis test set up to show that the sample parameter would be higher than the population parameter.

Example

Let X: the performance of Apple Inc. stock

μ : the average performance of Apple Inc. stock

$H_0: \mu \leq 13.5$ Against $H_1: \mu > 13.5$



Summary

Example	Hypothesis	Test Type
The company manager claims that the average weight of cookies in a gift box is 235 gm	$H_0: \mu = 235$ $H_1: \mu \neq 235$	Two Tailed Test
The station officer claims that the bullet train takes less than 2 hours from London to Brussels	$H_0: \mu \geq 2$ $H_1: \mu < 2$	Left Tailed Test
To test whether the average number of boat rides are more than 10 per day	$H_0: \mu \leq 10$ $H_1: \mu > 10$	Right Tailed Test

Decision Making Methods

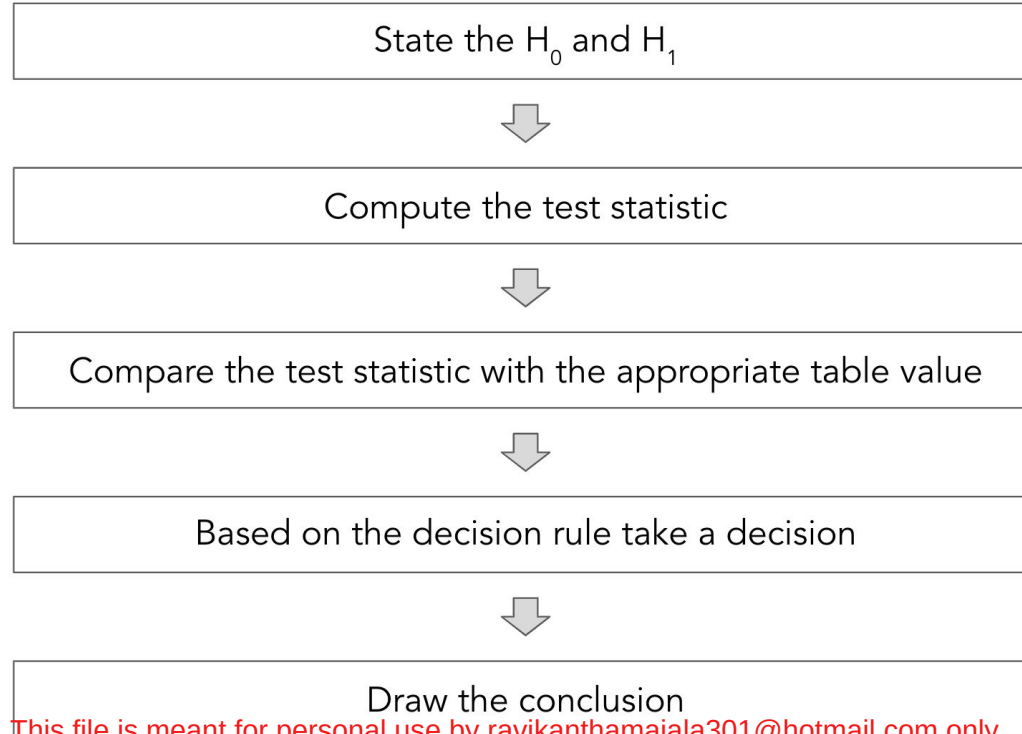
Decision making methods

- Test statistic, p-value and confidence interval are the three different criterias to test the acceptance/ rejection of null hypothesis
- Test statistic is a sample statistic that is used to conduct a hypothesis test
- The p-value criteria returns the probability of getting the value greater than or equal to the significance level
- The null hypothesis is accepted if the confidence interval contains the specified population parameter

Test statistic

- The decision whether to accept or reject the null hypothesis is based on the test statistic value
- Depending upon the problem in hand the test statistic changes
- The test statistic is used to quantify the sample data that distinguishes the null and alternative hypothesis
- The sampling distribution of a test statistic under the null hypothesis is used to calculate the p-value

Procedure - using test statistic



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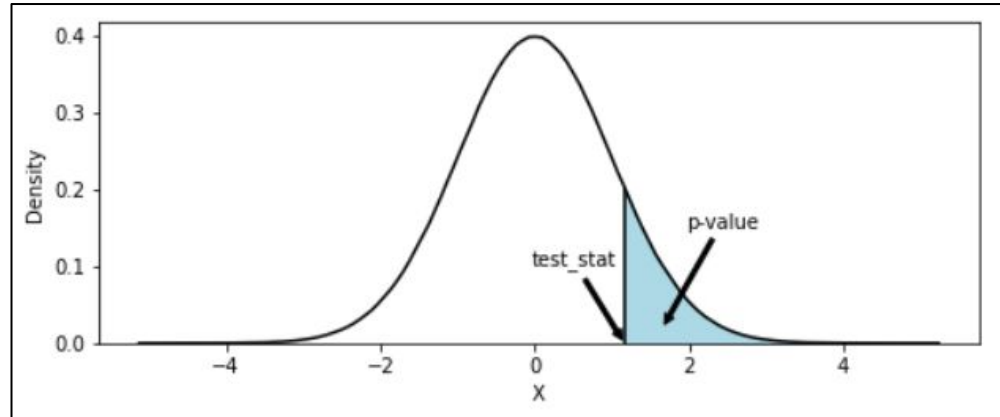
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The p-value

- The p-value is the mostly used criteria while performing the hypothesis test
- It is the probability of getting a value greater than or equal to the test statistic
- If T is the **test statistic** and T_0 is calculated value then observed level of significance or p-value is $P(T > T_0 | H_0 \text{ is true})$
- The test is significant if the p-value is less than or equal to the level of significance (α)

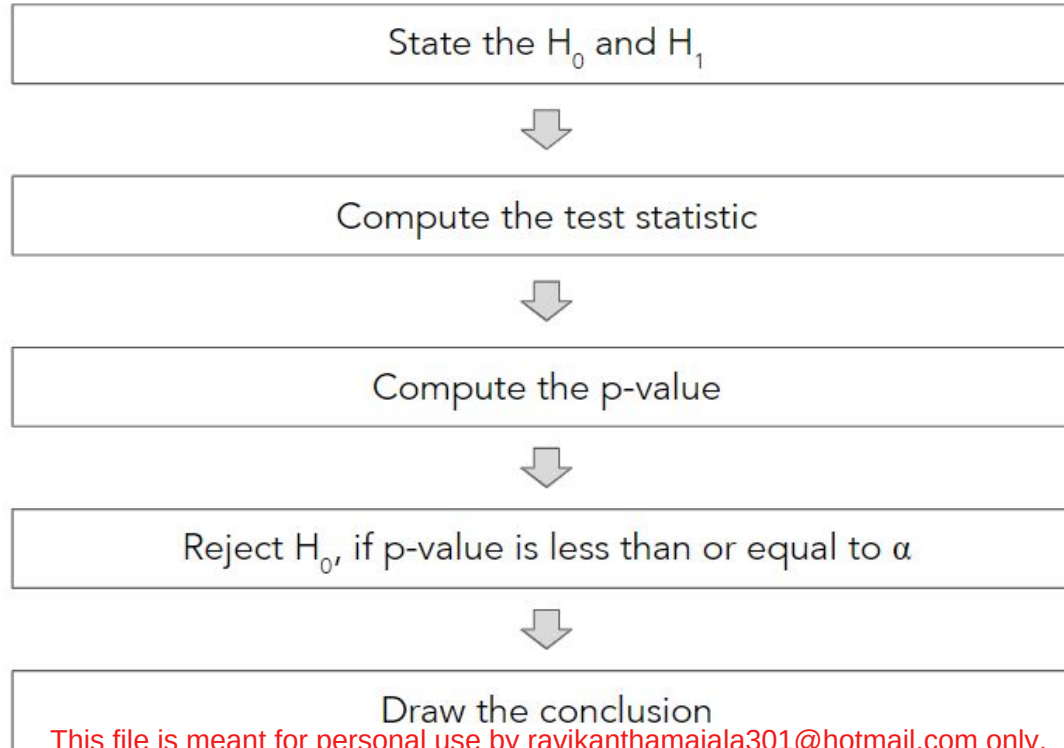
The p-value

- The smaller p-value supports the alternative hypothesis, as it exhibits the difference between the observed data and the null hypothesis



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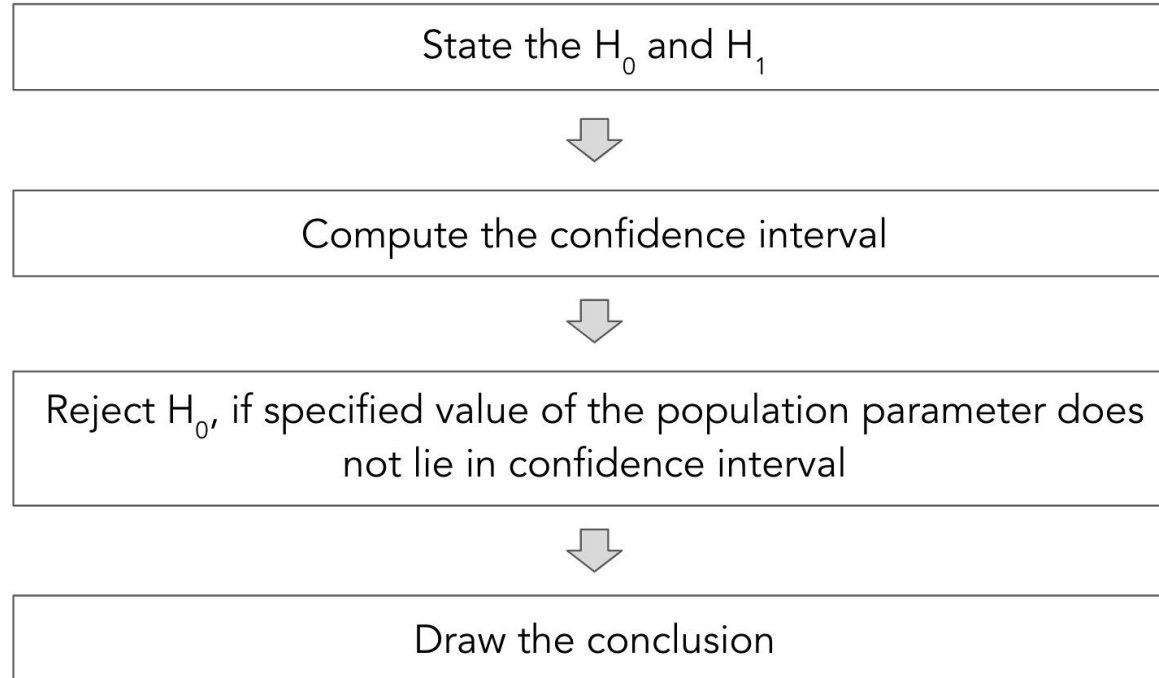
Procedure - using p-value



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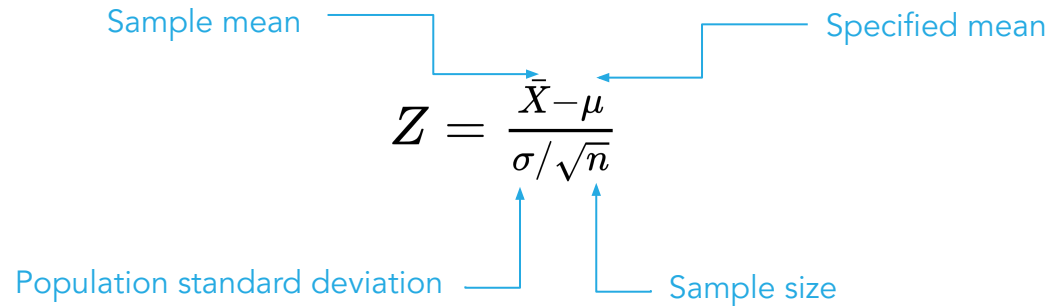
Procedure - using confidence interval



Tests based on Z statistic

Tests based on Z statistic

- The test statistic Z is given by


$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

- Under H_0 , the test statistic follows normal distribution
- It is used to test whether the population mean is equal to a specific mean (μ)

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The python code to conduct a Z test for population mean is

```
statsmodels.stats.weightstats.ztest(sample, value, alternative)
```

Decision rule

	H_1	Based on critical region	Based on p-value	Based on confidence interval
For two tailed test	$\mu \neq \mu_0$	Reject H_0 if $ Z > Z_{\alpha/2}$	Reject H_0 if p-value is less than or equal to the level of significance	Reject H_0 if μ_0 does not lie in the confidence interval
For left tailed test	$\mu < \mu_0$	Reject H_0 if $Z < -Z_\alpha$		
For right tailed test	$\mu > \mu_0$	Reject H_0 if $Z > Z_\alpha$		



Two tailed test

Question:

A car manufacturing company claims that the mileage of their new car is 25 kmph with a standard deviation of 2.5 kmph. A random sample of 45 cars was drawn and recorded their mileage as per the standard procedure. From the sample, the mean mileage was seen to be 24 kmph. Is this evidence to claim that the mean mileage is different from 25kmph? (assume normality of data)

Test the claim using critical region technique, p-value technique and confidence interval technique. [Use $\alpha = 0.01$].



Two tailed test

Solution:

A car manufacturing company claims that the mileage of a car is 25 kmph with a standard deviation of 2.5 kmph.

Thus $\mu = 25$ kmph and $\sigma = 2.5$ kmph

A random sample of 45 cars was drawn and recorded their mileage as per the standard procedure. From the sample, the mean mileage was seen to be 24 kmph. $\bar{X}$

Thus $\bar{X} = 24$ kmph and $n = 45$

To find: Evidence to claim that the mean mileage is different from 25kmph

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Two tailed test

Solution:

Here X: the mileage of the new car

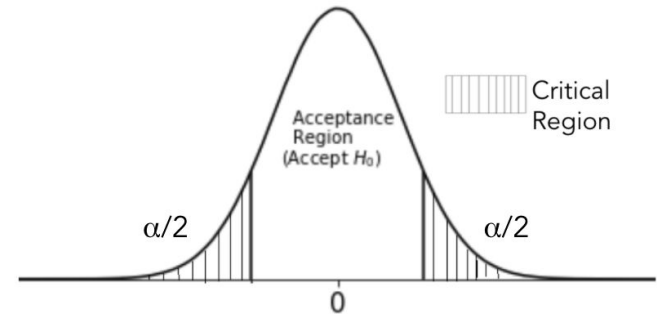
$$X \sim N(\mu, \sigma^2)$$

To test, $H_0 : \mu = 25$ against $H_1 : \mu \neq 25$

Obtain the solution using critical region technique.

The test statistics:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$



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Two tailed test

Solution:

The test statistics:

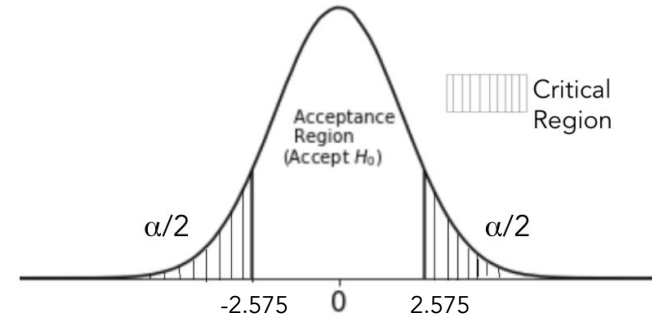
$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{24 - 25}{2.5 / \sqrt{45}} = -2.68$$

Here $\alpha = 0.01$, so $z_{0.005} = 2.575$

Decision: Reject H_0 , if $|Z_{calc}| > Z_{\alpha/2}$. Here $|-2.68| > 2.575$.

Thus reject H_0 .

Thus there is enough evidence to conclude that the mean mileage is different from 25 kmph.





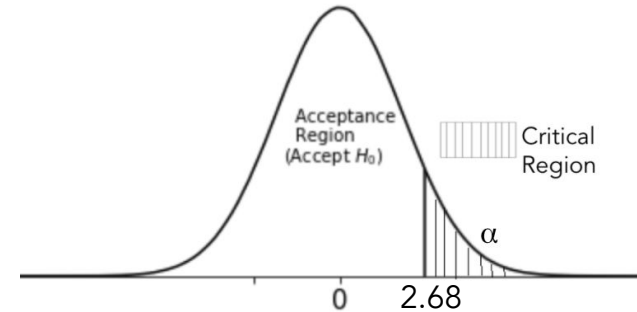
Two tailed test

Solution:

Let us obtain the decision using the p-value method.

$$Z_{calc} = -2.68$$

$$\begin{aligned}\text{The p-value} &= 2 \cdot P(|Z| > |Z_{calc}|, \text{ under } H_0) \\ &= 2 \cdot P(|Z| > 2.68, \mu = 25) \quad \dots \text{(due to symmetry)} \\ &= 0.0074\end{aligned}$$



Since p-value < 0.01, we reject H_0 .

Thus there is enough evidence to conclude that the mean mileage is different from 25 kmph.

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Two tailed test

Solution:

Solution using confidence interval method.

The confidence interval is given by $\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

$$\begin{aligned}\text{The CI} &= \left(\bar{X} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right) = \left(24 - 2.575 \frac{2.5}{\sqrt{45}}, 24 + 2.575 \frac{2.5}{\sqrt{45}} \right) \\ &= (23.05, 24.95)\end{aligned}$$

Since $\mu_0 = 25$, does not lie in $(23.05, 24.95)$, we reject H_0 .

Thus there is enough evidence to conclude that the mean mileage is different from 25 kmph.

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Two tailed test

Python solution: Calculate Z_{calc}

```
# define a function to calculate the Z-test statistic
# pass the population mean, population standard deviation, sample size and sample mean
def z_test(pop_mean, pop_std, n, samp_mean):

    # calculate the test statistic
    z_score = (samp_mean - pop_mean) / (pop_std / np.sqrt(n))

    # return the z-test value
    return z_score

# calculate the test statistic using the function 'z_test'
z_score = z_test(pop_mean = 25, pop_std = 2.5, n = 45, samp_mean = 24)
print("Z-score:", z_score)

Z-score: -2.6832815729997477
```

Reject H_0 , as $|Z_{\text{calc}}| > Z_{\alpha/2}$. Here $|-2.68| > 2.575$

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Two tailed test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'cdf()' to calculate  $P(Z \leq z\_score)$ 
p_value = stats.norm.cdf(z_score)

# for a two-tailed test multiply the p-value by 2
req_p = p_value*2
print('p-value:', req_p)

p-value: 0.007290358091535638
```

Since $p\text{-value} < 0.01$, we reject H_0 .



Two tailed test

Python solution: Calculate 99% confidence interval

```
# calculate the 99% confidence interval for the population mean
# pass the sample mean to the parameter, 'loc'
# pass the scaling factor (pop_std / n^(1/2)) to the parameter, 'scale'
print('Confidence interval:', stats.norm.interval(0.99, loc = samp_mean, scale = pop_std / np.sqrt(n)))
```

Confidence interval: (23.040045096471452, 24.959954903528548)

Since $\mu_0 = 25$, does not lie in the confidence interval, we reject H_0 .



Left tailed test

Question:

A sample of 900 PVC pipes is found to have an average thickness of 12.5 mm. Can we assume that the sample is coming from a normal population with mean 13mm against that it is less than 13 mm. The population standard deviation is 1 mm. Test the hypothesis using the p-value method at 5% level of significance.



Left tailed test

Solution:

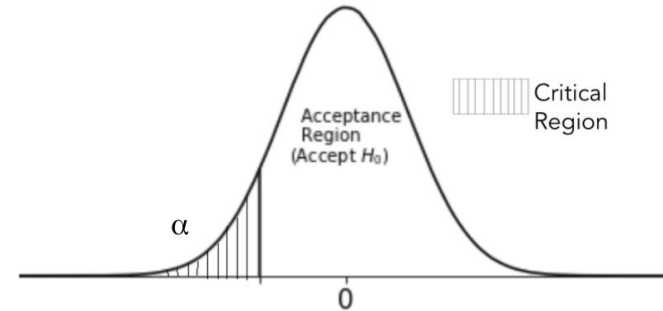
Here X: the thickness of PVC pipes

$$X \sim N(\mu, \sigma^2)$$

Here, $\mu = 13\text{mm}$, $\sigma = 1\text{ mm}$, $\bar{X} = 12.5$, $n = 900$

To test, $H_0 : \mu \geq 13$ against $H_1 : \mu < 13$

Obtain the solution using p-value technique.





Left tailed test

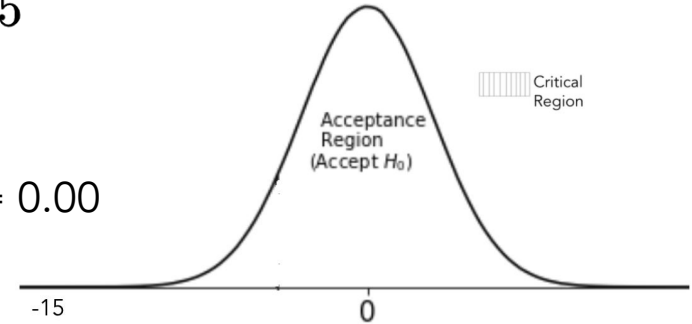
Solution:

The test statistics:
$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{12.5 - 13}{1 / \sqrt{900}} = -15$$

The p-value = $P(Z < Z_{calc}, \text{ under } H_0) = P(Z < -15, \mu = 13) = 0.00$

Since p-value < 0.05, we reject H_0 .

Thus there is enough evidence to conclude that the sample may not be regarded as coming from population with mean 13





Left tailed test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic  
# use 'cdf()' to calculate  $P(Z \leq -15)$   
p_value = stats.norm.cdf(-15)  
  
print('p-value:', p_value)  
  
p-value: 3.6709661993126986e-51
```

Since $p\text{-value} < 0.05$, we reject H_0 .



Right tailed test

Question:

An e-commerce company claims that the mean delivery time of food items on their website in NYC is 60 minutes with a standard deviation of 30 minutes. A random sample of 45 customers ordered from the website, and the mean time for delivery was found to be 75 minutes. Is this enough evidence to claim that the mean time to get items delivered is more than 60 minutes. (assume normality of data)

Test the claim using p-value technique. [Use $\alpha = 0.05$].



Right tailed test

Solution:

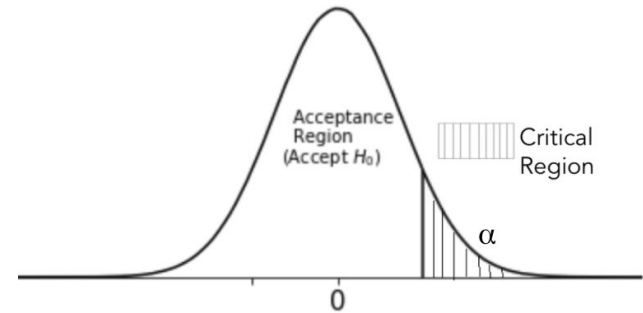
Here X: Time (in minutes) to receive the delivery

$$X \sim N(\mu, \sigma^2)$$

Here, $\mu = 60$, $\sigma = 30$, $\bar{X} = 75$, $n = 45$

To test, $H_0 : \mu \leq 60$ against $H_1 : \mu > 60$

Obtain the solution using p-value technique.





Right tailed test

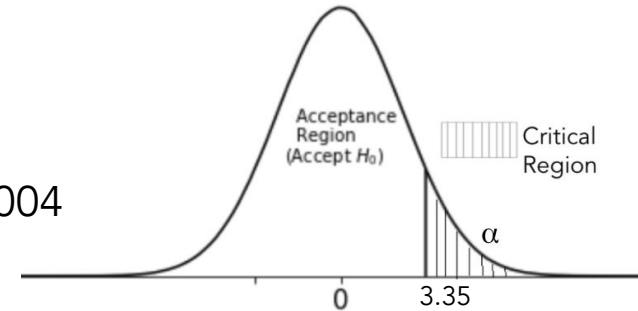
Solution:

The test statistics:
$$Z_{calc} = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} = \frac{75 - 60}{30 / \sqrt{45}} = 3.35$$

The p-value = $P(Z > Z_{calc}, \text{ under } H_0) = P(Z > 3.35, \mu = 60) = 0.0004$

Since p-value < 0.05, we reject H_0 .

Thus there is enough evidence to conclude that the mean delivery time is more than 60 min





Right tailed test

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'sf()' to calculate  $P(Z > 3.35)$ 
p_value = stats.norm.sf(3.35)

print('p-value:', p_value)

p-value: 0.0004040578018640207
```

Since $p\text{-value} < 0.05$, we reject H_0 .

Tests based on Z statistic

Assumptions:

- The data is continuous in nature
- The sample is assumed to be drawn from a population following the normal distribution
- The sample is a simple random sample (each observation is equally likely to be drawn)



Test for normality

- The hypothesis to test whether the population is normally distributed

H_0 : The data is normal

against

H_1 : The data is not normal

- Failing to reject H_0 , implies that the data is normally distributed
- The Shapiro-Wilk test is used

Note: The python code for Shapiro test is `scipy.stats.shapiro(Sample)`

Summary

- The Z test can be used if the sample data is drawn from a normally distributed population
- The normality of the dataset can be checked using Shapiro-Wilk test
- Two tailed as well as one tailed (left/ right) Z test can be performed
- The test statistic for the Z test follows normal distribution under the null hypothesis
- The test statistic, p-value and confidence interval can be used to analyze the significance of the test

Thank You
Happy Learning :)